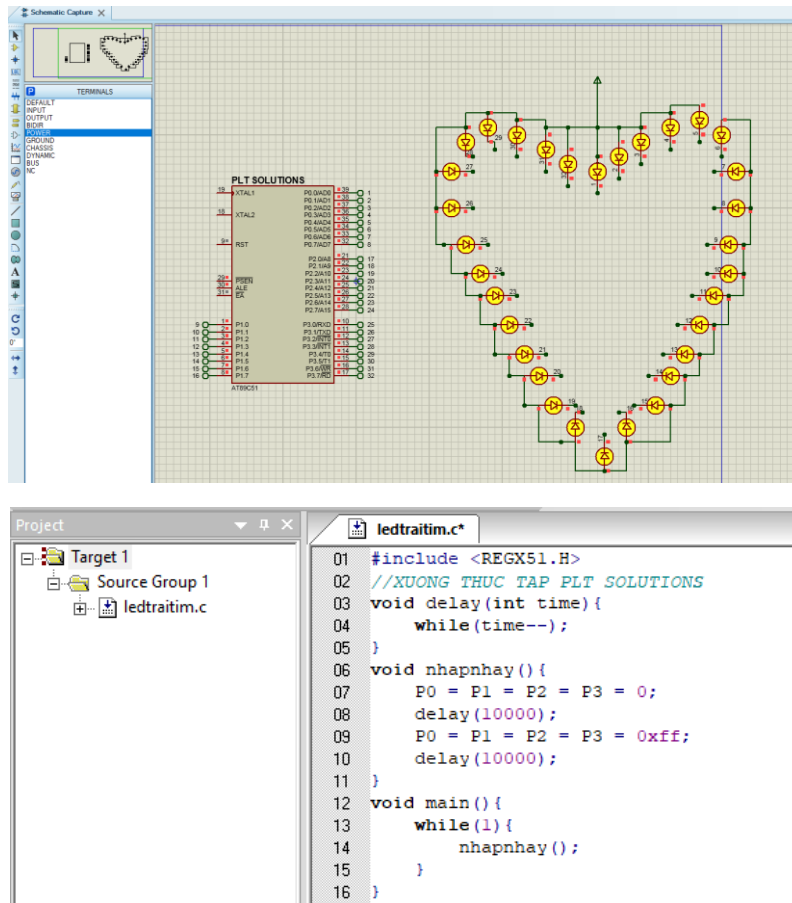
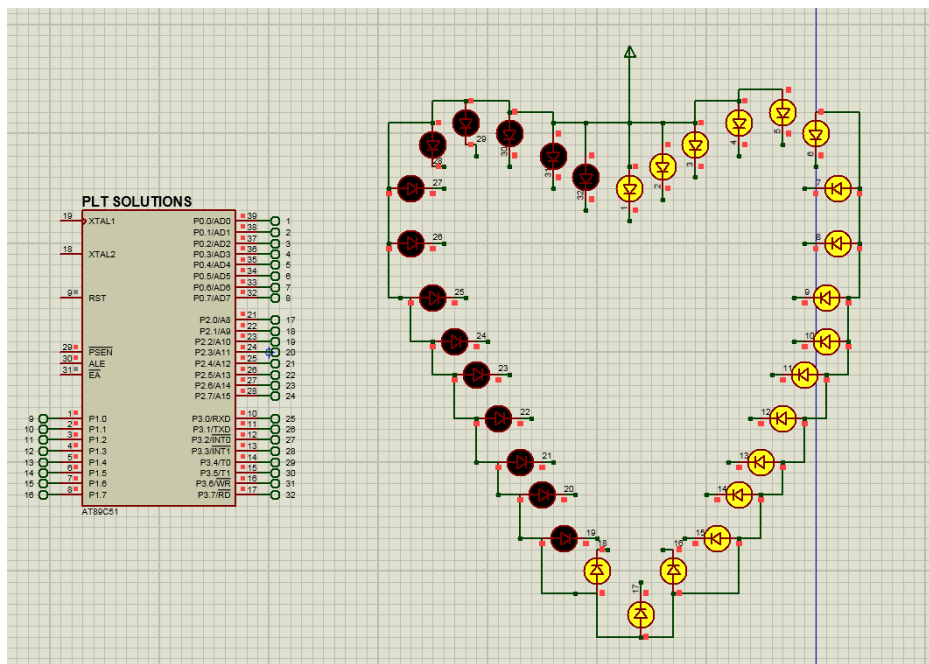


LAB 9 – LED TRÁI TIM

1. Hiệu ứng 32 bóng LED nhấp nháy:



2. Hiệu ứng tăng dần - 32 led được bật lên theo chiều thuận kim đồng hồ



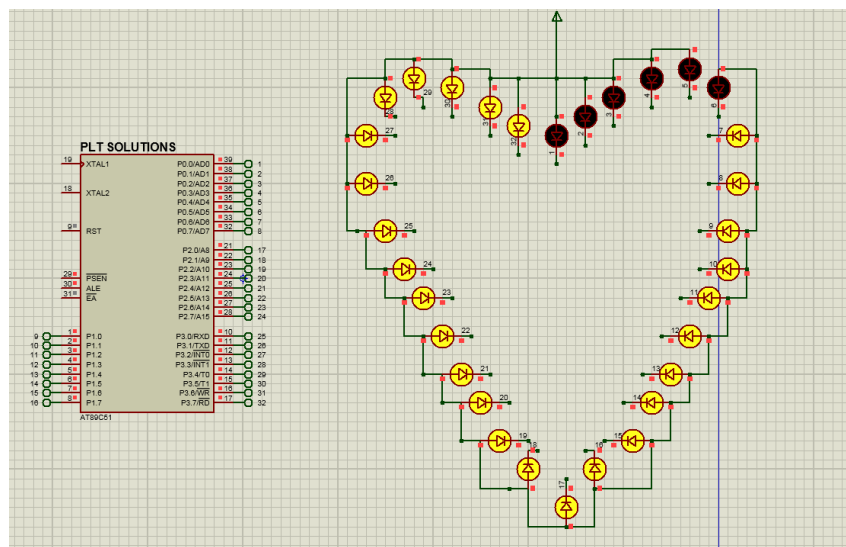
```

Project Target 1
Source Group 1
ledtraitim.c

ledtraitim.c
01 #include <REG51.H>
02 unsigned int i, j, k;
03
04 void delay(int time){
05     while(time--);
06 }
07 //XUONG THUC TAP PLT SOLUTIONS
08 void sangdanledthuan(){
09     unsigned char ThuanChieu[]={0xfe, 0xfc, 0xf8, 0xf0, 0xe0, 0xc0, 0x80, 0x00};
10     unsigned char NguocChieu[]={0x7f, 0x3f, 0x1f, 0x0f, 0x07, 0x03, 0x01, 0x00};
11     P0 = P1 = P2 = P3 = 0xff;
12
13     for(i = 0; i < 8; i++){
14         P0 = ThuanChieu[i];
15         delay(10000);
16     }
17     delay(10000);
18
19     for(i = 0; i < 8; i++){
20         P1 = ThuanChieu[i];
21         delay(10000);
22     }
23     delay(10000);
24
25     for(i = 0; i < 8; i++){
26         P2 = ThuanChieu[i];
27         delay(10000);
28     }
29     delay(10000);
30
31     for(i = 0; i < 8; i++){
32         P3 = ThuanChieu[i];
33         delay(10000);
34     }
35     delay(10000);
36 }
37
38 // Sang dan het cac Led theo chieu thuan kim dong ho
39 void main(){
40     while(1){
41         sangdanledthuan();
42     }
43 }

```

3. Hiệu ứng tăng dần - 32 led được bật lên theo chiều ngược kim đồng hồ



```

Project Target 1
Source Group 1
ledtraitim.c

ledtraitim.c
01 #include <REG51.H>
02 unsigned int i, j, k;
03
04 void delay(int time){
05     while(time--);
06 }
07 //XUONG THUC TAP PLT SOLUTIONS
08 void sangdanlednguoc(){
09     unsigned char ThuanChieu[]={0xfe, 0xfc, 0xf8, 0xf0, 0xe0, 0xc0, 0x80, 0x00};
10     unsigned char NguocChieu[]={0x7f, 0x3f, 0x1f, 0x0f, 0x07, 0x03, 0x01, 0x00};
11     P0 = P1 = P2 = P3 = 0xff;
12
13     for(i = 0; i < 8; i++){
14         P3 = NguocChieu[i];
15         delay(10000);
16     }
17     delay(10000);
18
19     for(i = 0; i < 8; i++){
20         P2 = NguocChieu[i];
21         delay(10000);
22     }
23     delay(10000);
24
25     for(i = 0; i < 8; i++){
26         P1 = NguocChieu[i];
27         delay(10000);
28     }
29     delay(10000);
30
31     for(i = 0; i < 8; i++){
32         P0 = NguocChieu[i];
33         delay(10000);
34     }
35     delay(10000);
36 }
37
38 // Sang dan het cac Led theo chieu nguoc kim dong ho
39 void main(){
40     while(1){
41         sangdanlednguoc();
42     }
43 }

```

4. Các bóng LED tắt dần theo chiều thuận kim đồng hồ

```
Project
Target 1
Source Group 1
ledtraitim.c

01 #include <REGX51.H>
02 unsigned int i, j, k;
03
04 void delay(int time){
05     while(time--);
06 }
07 //XUONG THUC TAP PLT SOLUTIONS
08 void tatdanledthuan(){
09     unsigned char ThuanChieu[]={0x01, 0x03, 0x07, 0x0F, 0x1F, 0x3F, 0x7F, 0xFF};
10     unsigned char NguocChieu[]={0x7F, 0x3F, 0x1F, 0x0F, 0x07, 0x03, 0x01, 0x00};
11     P0 = P1 = P2 = P3 = 0x00;
12
13     for(i = 0; i < 8; i++){
14         P0 = ThuanChieu[i];
15         delay(10000);
16     }
17     delay(10000);
18
19     for(i = 0; i < 8; i++){
20         P1 = ThuanChieu[i];
21         delay(10000);
22     }
23     delay(10000);
24
25     for(i = 0; i < 8; i++){
26         P2 = ThuanChieu[i];
27         delay(10000);
28     }
29     delay(10000);
30
31     for(i = 0; i < 8; i++){
32         P3 = ThuanChieu[i];
33         delay(10000);
34     }
35     delay(10000);
36 }
37
38 // Tat dan cac Led theo chieu thuan kim dong ho
39 void main(){
40     while(1){
41         tatdanledthuan();
42     }
43 }
```

5. Các bóng LED tắt dần theo chiều ngược kim đồng hồ

```
Project
Target 1
Source Group 1
ledtraitim.c

01 #include <REGX51.H>
02 unsigned int i, j, k;
03
04 void delay(int time){
05     while(time--);
06 }
07 //XUONG THUC TAP PLT SOLUTIONS
08 void tatdanlednguoc(){
09     unsigned char ThuanChieu[]={0x01, 0x03, 0x07, 0x0F, 0x1F, 0x3F, 0x7F, 0xFF};
10     unsigned char NguocChieu[]={0x80, 0xC0, 0xE0, 0xF0, 0xFB, 0xFC, 0xFE, 0xFF};
11     P0 = P1 = P2 = P3 = 0x00;
12
13     for(i = 0; i < 8; i++){
14         P3 = NguocChieu[i];
15         delay(10000);
16     }
17     delay(10000);
18
19     for(i = 0; i < 8; i++){
20         P2 = NguocChieu[i];
21         delay(10000);
22     }
23     delay(10000);
24
25     for(i = 0; i < 8; i++){
26         P1 = NguocChieu[i];
27         delay(10000);
28     }
29     delay(10000);
30
31     for(i = 0; i < 8; i++){
32         P0 = NguocChieu[i];
33         delay(10000);
34     }
35     delay(10000);
36 }
37
38 // Tat dan cac Led theo chieu nguoc kim dong ho
39 void main(){
40     while(1){
41         tatdanlednguoc();
42     }
43 }
```

XUÔNG THỰC TẬP PLT SOLUTIONS

<https://pltpro.net>

<https://xaydungphanmem.com>

www.youtube.com/@pltlsolutions3010